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$$A=\begin{pmatrix}3 & 4 & 7\\ 2 & 1 & 3\\ 7 & 2 & 1\end{pmatrix}$$

Solution:

Determinant of the given matrix is,

$$|A|=\begin{vmatrix}3 & 4 & 7\\ 2 & 1 & 3\\ 7 & 2 & 1\end{vmatrix}=3\begin{vmatrix}1 & 3\\ 2 & 1\end{vmatrix}-4\begin{vmatrix}2 & 3\\ 7 & 1\end{vmatrix}+7\begin{vmatrix}2 & 1\\ 7 & 2\end{vmatrix}$$

• 3 (1 • 6) • 4 (2 • 21)
• 7 (4 • 7)

• 3 (• 5) • 4 (• 19) • 7 (• 3)

• • 15 • 76 • 21

$$\boxed{|A| \bullet 40}$$

The value of determinant is 40.

12.3.3 Cramer's Rule



(1704-1752)

Cramer's rule provides the solution of a system of linear equations with '**n**' variables and '**n**' equations. It helpsto arrive at a unique solution of a system of linear equations with as many equations as unknowns.

If the given equations are

$$a_{11}x \bullet a_{12}y \bullet a_{13}z \bullet b_1$$

$$a_{21}x \bullet a_{22}y \bullet a_{23}z \bullet b_2$$

$$a_{31}x \bullet a_{32}y \bullet a_{33}z \bullet b_3$$

then

$$x=\frac{\Delta x}{\Delta},\quad y=\frac{\Delta y}{\Delta},\quad z=\frac{\Delta z}{\Delta}$$

where, $\Delta=\begin{vmatrix}a_{11} & a_{12} & a_{13}\\ a_{21} & a_{22} & a_{23}\\ a_{31} & a_{32} & a_{33}\end{vmatrix}$

$$\Delta_x=\begin{vmatrix}b_1 & a_{12} & a_{13}\\ b_2 & a_{22} & a_{23}\\ b_3 & a_{32} & a_{33}\end{vmatrix},\Delta_y=\begin{vmatrix}a_{11} & b_1 & a_{13}\\ a_{21} & b_2 & a_{23}\\ a_{31} & b_3 & a_{33}\end{vmatrix},$$

$$\Delta z=\begin{vmatrix}a_{11} & a_{12} & b_1\\ a_{21} & a_{22} & b_2\\ a_{31} & a_{32} & b_3\end{vmatrix}$$

Example: 12.8

Find the value of x and y in the equations by using Cramer's rule. x • 3y • 1 and 3x • 2y • 14

Key Note

If the determinant $\Delta=0$, then solution does not exist.

Solution:

Given equations are

x • 3y • 1
3x • 2y • 14

Then the equations in the matrix form :

$$\begin{pmatrix}1 & 3\\ 3 & -2\end{pmatrix}\begin{pmatrix}x\\ y\end{pmatrix}=\begin{pmatrix}1\\ 14\end{pmatrix}$$

Calculating Δ, $\Delta=\begin{vmatrix}1 & 3\\ 3 & -2\end{vmatrix}$

• • 2 • 9
• • 11

$\Delta \neq 0$, Hence solution exists.

$$\Delta_x=\begin{vmatrix}1 & 3\\ 14 & -2\end{vmatrix}=-2-42 \bullet \bullet 44$$

$$\Delta_y=\begin{vmatrix}1 & 1\\ 3 & 14\end{vmatrix}=14-3=11$$